Constellation

10 CFR 50.73

LG-25-118

August 12, 2025

U.S. Nuclear Regulatory Commission
ATTN: Document Control Desk
Washington, DC 20555-0001

Limerick Generating Station, Unit 1
Renewed Facility Operating License No. NPF-39
NRC Docket No. 50-352

Subject: LER 2025-002-00 Loss of Power to the 1A Reactor Protection System Resulted in Multiple Invalid Primary Containment Isolations

In accordance with the requirements of 10 CFR 50.73(a)(2)(iv)(A), Limerick Generating Station hereby submits the enclosed Licensee Event Report.

There are no commitments contained in this letter.

If you have any questions, please contact Jordan Rajan at (610) 718-3400.

Respectfully,

Michael F. Gillin

Michael F. Gillin
Vice President – Limerick Generating Station
Constellation Energy Generation, LLC

cc: Administrator Region I, USNRC
USNRC Senior Resident Inspector, Limerick Generating Station

NRC FORM 366 (04-02-2024) U.S. NUCLEAR REGULATORY COMMISSION

LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/)

1. Facility Name		2. Docket Number	3. Page
Limerick Generating Station, Unit 1	■ 050 / ☐ 052	352	1 OF 3

4. Title

Loss of Power to the 1A Reactor Protection System Resulted in Multiple Invalid Primary Containment Isolations

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
06	13	2025	2025	- 002 -	00	08	12	2025	Facility Name	☐ 052	Docket Number

9. Operating Mode	10. Power Level
1	100

11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: *(Check all that apply)*

10 CFR Part 20	☐ 20.2203(a)(2)(vi)	10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☑ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		

☐ **OTHER** (Specify here, in abstract, or NRC 366A).

12. Licensee Contact for this LER

Licensee Contact	Phone Number (Include area code)
Jordan Rajan, Regulatory Assurance Manager	610-718-3400

13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
B	JC	RLY	A500	Yes					

14. Supplemental Report Expected		15. Expected Submission Date	Month	Day	Year
☑ No	☐ Yes (If yes, complete 15. Expected Submission Date)				

16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On June 13, 2025, at 10:23, Limerick Generating Station (LGS) Unit 1 experienced a loss of power to the "A" Reactor Protection System (RPS) 120 VAC Distribution Panel that resulted in a Unit 1 "A" RPS half scram, Main Steam Isolation Valve (MSIV) half isolation signal, and primary containment isolations in multiple systems. The most probable cause of the power loss was due to an under-frequency relay spurious trip condition. The under-frequency relay was replaced and post maintenance testing was completed satisfactorily. Power was restored to the panel and all systems were returned to service.

This event caused invalid primary containment isolation signals in multiple systems which is reportable in accordance with 10 CFR 50.73(a)(2)(iv)(A).

1. FACILITY NAME		2. DOCKET NUMBER	3. LER NUMBER YEAR	SEQUENTIAL NUMBER	REV NO.
Limerick Generating Station, Unit 1	■ 050 / ☐ 052	352	2025	002	00

NARRATIVE

Unit Condition Prior to Event

Prior to the event, Unit 1 was in Operational Condition (OPCON) 1 (Power Operation) at approximately 100 percent power. There were no other structures, systems, or components out of service that contributed to this event.

Description of the Event

On June 13, 2025, at 10:23, Limerick Generating Station (LGS) Unit 1 was at 100 percent power. The Unit 1 "A" (1A) Reactor Protection System (RPS) [JC] 120 VAC Distribution Panel experienced a loss of power resulting in a 1A RPS half scram, Main Steam Isolation Valve (MSIV) half isolation signal, and multiple primary containment isolations. The following systems isolated: Drywell Chilled Water (DWCW), Reactor Enclosure Cooling Water (RECW), Primary Containment Instrument Gas (PCIG), Reactor Water Cleanup (RWCU), Containment Leak Detection, Drywell Equipment & Floor Sump Drains, and Reactor Enclosure ventilation. A single train of the Standby Gas Treatment System (SGTS) started. All systems responded as designed.

Troubleshooting determined the most probable cause to be a spurious trip of an under-frequency relay, 81-CY24801, for a supply circuit breaker to the 1A RPS 120 VAC Distribution Panel. The 1A RPS circuit breaker bypass disconnect switch was closed to restore power to the panel and reset isolation signals. On June 14, 2025, the relay was replaced and the normal alignment was restored.

These system actuations were not in response to actual plant conditions or parameters satisfying the requirements for initiation of the system. The plant responded as expected due to loss of power to the 1A RPS 120 VAC Distribution Panel. This event is reportable as an invalid actuation in accordance with 10 CFR 50.73(a)(2)(iv)(A).

Analysis of the Event

The supply circuit breaker within the 1A RPS 120 VAC Distribution Panel tripped open, due to a spurious actuation of the under-frequency relay, 81-CY24801. System actuations responded to the loss of power condition as expected with no issues. The station performed troubleshooting to identify the cause of the loss of power to the 1A RPS 120 VAC Distribution Panel. Once the problem was identified, the station restored power to the panel, reset the isolations, and returned the systems to service.

Safety Consequence

There were no actual nuclear safety consequences. The affected systems responded as designed to an invalid under-frequency trip signal. No safety functions were lost during this event.

Cause of the Event

The most probable cause of the loss of power to the 1A RPS 120 VAC Distribution Panel was due to a spurious trip of the under-frequency relay. Failure analysis was unable to determine a definitive cause for the spurious operation.

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NARRATIVE

Corrective Actions Completed

The under-frequency relay was replaced.

Corrective Actions Planned

The station will evaluate implementing a replacement strategy for the under-frequency relay.

Previous Similar Occurrences

Unit 1 LER 2024-002-00 Invalid Actuation Due to a Trip of 1B RPS Circuit Breaker - This trip was caused by a defective circuit breaker occurring after the completion of testing.

Component Data

System - JC (Plant Protection System)
Component - RLY
Manufacturer - A500 (Asea Inc)